

Investment Outlook and Guidance for The Senior Care Industry

Kelly R. Simmons

Northwest Missouri State University, Maryville MO 64468, USA
S557075@nwmissouri.edu

Abstract. This project examines various metrics related to healthcare facilities across different states in the United States based on the provided dataset. The data covers information on the number of facilities, monthly and annual metrics, staffing details, population statistics, deficiencies, fines, and other relevant indicators. The analysis sheds light on the healthcare landscape, revealing variations and trends in facility sizes, workforce, financial performance, and regulatory compliance. The goal of the project is to provide guidance to which states would be best for investment into the Nursing home care industry and which states should be avoided.

Keywords: Performance metrics, Dataset analysis, Workforce trends, Financial performance, Gross patient revenue, Facility capacity

1 Introduction

As the nation ages and the length of life increases, the senior care industry, also known as the skilled nursing facility industry, has quietly been a staple in the portfolios of many investors. Even through the recent shock of Covid and the always present staffing shortages, Investors, especially private equity players, see increasing value in nursing home portfolios, according to the National Investment Center for Seniors and Care. [2] Can the skilled nursing facility not only recover but also grow in the future? Should this industry continue to be the backbone of investment portfolios? This project will use several datasets including, Total Number of Certified Nursing Facilities excel file, Nursing Home Cost Per State excel file, Quarterly Staffing Statistics by State excel files, and State by State Violations 2023 excel file, to see if this is the case. The goal of this project is to provide an in-depth look at the industry and its future. What areas are the best to invest in. For example, which states or cities would be best to invest in. This report will look at population demographics to determine assumed growth in areas while considering market factors that may negatively or positively influence the given market. Industry reports on the skilled nursing facility industry can be found everywhere. This report will attempt to provide specific state information to better target the right areas to invest in.

1.1 The report is organized in the following manner: Phase one presents the industry background. It is important to understand how the data is provided

in the way it is. For example, nursing homes costs are split into three areas. Room cost and care costs. These will not be the same for each patient. Knowing this, averages may need to be instituted into the final calculation to provide assumptions on future growth. Phase two presents the datasets used and how they will be cleaned to provide accurate outputs. The cleaning process portion will be organized into three steps. Phase three will be the results. This will present a summary of what the data found and an explanation how the user can use the information. Phase four is the conclusion. This will present the use with an overall use of the information.

1.1 Industry Background

Before the data gathered can be used, it is important to first understand why the data is provided in the manner it is and how it relates to the industry. The first step is understanding what a nursing home facility is and how a nursing home is set up as far as it relates to cost. What is a nursing home facility? A nursing home is a facility for the residential care of older people, senior citizens, or disabled people. [3] How is a facility set up? The first category is housing. The number of rooms varies. There is no state or federal regulation that requires a certain number of rooms. However, the average number of certified nursing facility beds in the United States in 2022 was 106.4. [1] In a facility, each room can be either private or semi-private meaning two people to a room. This cost covers the room or portion of the room and meals for the day. The next cost is the care. Depending on what a patient needs, care can be provided by either in-house providers or from a third party. For example, a person recovering from a stroke may need physical therapy. This care is an extra cost. For this report, care cost will not be used as it is too wide ranging depending on the needs of the patient. There is no way to effectively capture that data set as the criteria would change from patient to patient. The cost for nursing home facilities is influenced by location like most other things. To give an example, the average cost of a private room in California would cost \$146,000 a year while the average cost for a private room in Kansas is \$81,760. An early assumption would be that the best place to invest would be in California as it charges more per patient. This figure does not consider the overhead cost of running the facility. This report answers those questions.

2 Data

This report utilizes datasets sourced from various entities, including The American Council on Aging, Grand View Research, Wealth Management.com, The Long-Term Care Coalition, and The Kaiser Family Foundation, all available in Excel formats. Consequently, there is no need for web scraping. In the Facility cost by State dataset, the pertinent attributes include State, Monthly, and Annual. The Quarterly Staffing dataset, also in Excel format, comprises five distinct datasets that will be amalgamated. However, only the State and Avg. Number of

Daily Worker attributes will be incorporated into this report. The State-by-State Violations dataset is another extensive Excel-format dataset, and the attributes selected for inclusion are State, Number of Homes, Total Deficiencies, Fines, and the Sum of Fines. Extraneous attributes that do not contribute to the overall report will be excluded. It is worth noting that an initial dataset detailing each state by its regions was excluded as the emphasis of the file was on regions rather than individual states. Below is a diagram of the data gathering process.

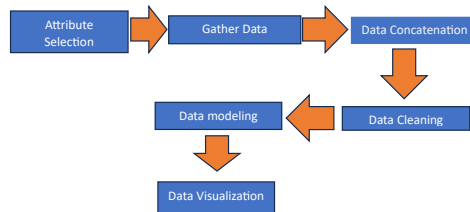


Fig. 1. Python code for the data gathering process.

3 Data Cleaning

Data cleaning is the most important aspect of any project. When working with large datasets, this task can quickly take up a lot of time. In this report, the datasets were for the most part gathered in either their Excel formats. For this reason, more complicated cleaning such as web scrapping and web mining and data streaming are not needed. This report consists of eight datasets. Getting the data to a usable state took the following steps.

3.1 Review the Data The datasets analyzed for this report underwent an initial high-level review to ensure they met the required criteria. This involved confirming the presence of the desired attributes for the report and assessing the overall formatting of the data. An illustrative instance of this quality check occurred with the Nursing Home Staffing by State dataset. Notably, before Q1 of 2021, the total count of nursing facility staff per state was recorded not by individual counts but by the number of hours worked. To maintain consistency across attributes, only data from Q4 of 2020 onwards was incorporated. Subsequently, the examination shifted to identifying any instances of missing data, and fortunately, none were

3.2 Data Cleaning The next step was to clean the data. This required a more in-depth look at the data. Below is a description of what cleaning was done to each dataset. The Number of Facilities by State dataset required no cleaning. It contained only two attributes, State and Number of Facilities. The final version of the Quarterly Staffing dataset is five separate datasets combined into one. Attributes such as State Abbreviation, Total Nurse HPRD, Total Rank, RN HPRD, and RN Rank were deleted from the dataset as the focus was on the number of staff, not the number of hours they worked or how that data ranked nationwide. The datasets included Washington DC. As a separate value. This input will be added to the state of Maryland. This left State, and Avg. Daily Census (Total). Once the unneeded attributes were removed, the datasets will be combined into one. The State-by-State Violations dataset is also a large dataset that is in an excel format. The attributes that will be used in this report are State, Number of Homes, Total Deficiencies, Fines and Sum of Fines. The remaining attributes: Homes, Serious Deficiencies, Infection Deficiencies, Staff Fully Vaccinated Against COVID-19, Average Fine, Payment Suspensions were removed as they do not contribute to the overall report. Washington DC. was included as its own value. It will be added to the state of Maryland. The U.S. territories of Guam and Puerto Rico were also included in this dataset. This report is concerned with the nursing home facilities in the United States. Guam and Puerto Rico were deleted from the Dataset. During the cleaning process VS Code was used. Python code was created to concatenate the data into a single dataset. It was discovered that due to each dataset having a state column the code created duplicates. To fix this Each state column in each dataset was organized alphabetically. They were then deleted except for one. The state column in Facility Cost by state was used as the organizer. All data was built off of that dataset. It was then run with the python code .

4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) played a key role in the project. The data used in the project comes from various sources. The EDA process allowed for the data to be compared individually and as it pertained to its data row. In the end, providing the results we wanted.

3.1 Preparation of the Data

Preparing the data analysis required took several steps. Designing the cleaning and reading the data into Python was the first task. The files had different formats in the data as well as the files themselves. making sure the data was compatible was the first hurdle. There were time when the results were formatted as percentages. This particular data had to be reformatted to their decimal form in an effort to make all data compatible. Once the results were gathered, a grading system had to be created. National average data was used as a baseline. Any totals falling on or close to the national average was given a grade of C. Data result that were above or below were given grades of A, B, D, or F.

3.2 Preliminary Findings

The initial findings did provide some surprises. Particularly in the in nursing home cost by state. States that were assumed would have a low cost actually had very high cost. Assumptions could be made as to why this is the case but that would include more of a cultural influence. Patterns in state by state staffing numbers showed the that staffing did not dip as much as assumed during the Covid pandemic. In fact in some states, the staffing number went up. Data in state population showed a some states have a larger than expected population. This would could lead to high transitions into nursing homes with in the state in the coming years. These trends are crucial to the investment outlooks for the nursing home industry. Key factors in investing in the nursing home industry are the the quantity of patients that will be entering the facilities in the coming years, staffing numbers. Is there a large or growing pool of qualified workers for the facilities. And cost, is it profitable to open a facility in certain areas depending on these variables.

-Data Collection, Integration, and Transformation: The first step was extracting needed data from the various excel files: Nursing home Cost by State, Facility Staffing Combined, Facility by State, Occ rate 2015 2023, Population stats, State by State Violations 2023 and Facility by State. In an effort to make the data easier to work with, data was taken from similar data sets and combined into one. It was then combined with the other files to make the universal file. Key attributes were then identified. This list included: State, Region, Private Room Annual Cost, Shared Annual Cost, Occupancy Rate Avg, Population 2023, Employee Average, Staff Increase/Decrease, Total Deficiencies, Fines, Sum of Fines, and Facility Total.

3.3 Python Analysis

Below is the code used in the data gathering and cleaning process.

Universal_code.py

```

1 import pandas as pd
2 import os
3
4 # Set the path to the directory containing your Excel files
5 input_directory = r'C:\Users\keros\.venv\Capstone-Project'
6
7 # Initialize a dictionary to store DataFrames for each file
8 dfs = {}
9
10 # List all files in the directory
11 all_files = os.listdir(input_directory)
12
13 # Specify desired columns for each file
14 desired_columns_nursing_home = ['State', 'Region', 'Private
    Room Annual Cost', 'Shared Room Annual Cost']
15
16 desired_columns_occ_rate = ['Occupancy Rate Avg']
17
18 desired_columns_population_stats = ['Population 2023']
19
20 desired_columns_facility_staffing_combined = ['Employee
    Average', 'Staff Increase/Decrease']
21
22 desired_columns_state_violations = ['Total Deficiencies', '
    Fines', 'Sum of Fines']
23
24 # Desired columns for Facility_by_state.csv
25 desired_columns_facility_by_state = ['Facility Total']
26
27 # Columns you want to include in the final DataFrame
28 desired_columns = ['State', 'Region', 'Private Room Annual
    Cost', 'Shared Room Annual Cost',
29                    'Employee Average', 'Staff Increase/
    Decrease', 'Population 2023',
30                    'Total Deficiencies', 'Fines', 'Sum of
    Fines', 'Facility Total']
31
32 # Loop through each file and read its data into the dictionary
33 for excel_file in all_files:
34     # Skip hidden files, directories, temporary Excel files,
    and non-CSV files
35     if excel_file.startswith('.') or os.path.isdir(os.path.
    join(input_directory, excel_file)) \
36         or excel_file.startswith('~$') or not excel_file.
    endswith('.xlsx'):
37         continue
38
39     excel_path = os.path.join(input_directory, excel_file)

```

```

40     try:
41         # Print columns before reading
42         actual_columns = pd.read_excel(excel_path, nrows=0).
columns
43         print(f"Actual Columns in {excel_file}: {
actual_columns}")
44
45         if excel_file == 'Facility_by_state.csv':
46             # For Facility_by_state.csv, use
desired_columns_facility_by_state
47             dfs[excel_file] = pd.read_csv(excel_path, usecols=
desired_columns_facility_by_state)
48         else:
49             # For other Excel files, use desired_columns
50             dfs[excel_file] = pd.read_excel(excel_path,
usecols=desired_columns)
51
52     except Exception as e:
53         print(f"Error reading {excel_file}: {e}")
54
55 # Concatenate all DataFrames in the dictionary
56 combined_data = pd.concat([df[desired_columns] for df in dfs.
values()], axis=1)
57
58 # Set the path for the output Excel file
59 output_file_path = r'C:\Users\keros\.venv\Capstone-Project\
outputdata.xlsx'
60
61 # Write the combined data to the output Excel file
62 combined_data.to_excel(output_file_path, index=False)
63
64 print(f"Data from {len(all_files)} files combined and saved to
{output_file_path}")

```

5 5. Model Selection

After executing the Universal code, we employ the Linear Regression Method to validate the regression of Gross Patient Revenue as the target variable against Monthly, Annual, Employee Average, Staff Increase/Decrease, Population, and Certified Beds by State. This method is chosen to illustrate the data trends, and it effectively handles missing values, including two rows with partial empty cells.

Anticipated outcomes from the Linear Regression model include the prediction of Gross Patient Revenue based on the specified features. Evaluation metrics such as Mean Squared Error (MSE) and R-squared (R²), accompanied by letter grades, offer both quantitative and qualitative assessments of the model's performance. The scatter plot further facilitates a visual examination of the model's predictions.

The objective is to develop a model that accurately captures relationships in the data and provides reliable predictions for Gross Patient Revenue. While the linear regression model demonstrates a reasonably good fit to the data, indicated by an R-squared grade of 'B,' there is room for improvement in predictive accuracy, as evidenced by the 'E' grade for Mean Squared Error. Further analysis and potential refinement of the model may enhance its overall performance.

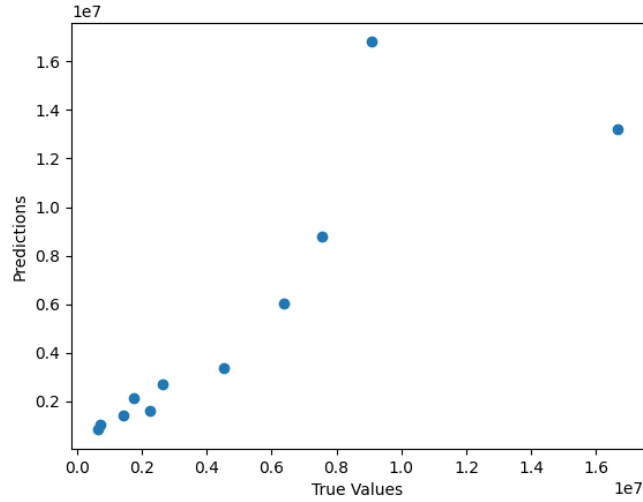


Fig. 2. Python code for Linear Regression Model test.

6 6. Results

The goal of the project was to gather state information on the nursing home facility industry and provide guidance as to which states would be the best locations for investing in the industry. Below are is the list of the top 10 best states for investments and the 10 worst states for investment.

6.1 Investment Score

The investment score is a quantitative measure devised to assess the desirability of investing in skilled nursing facilities across different states. It is calculated based on a combination of key factors that contribute to the overall attractiveness of an investment opportunity in this industry.

Calculation Methodology:

Weighted Criteria: The score considers various criteria, each assigned a specific weight based on its perceived significance in influencing investment success.

Population Demographics: The population demographic of each state plays a crucial role. States with a higher proportion of elderly individuals are often considered more attractive for senior care investments.

Financial Metrics: Financial metrics such as annual revenue, fines, and total deficiencies provide insights into the economic performance and regulatory compliance of the facilities. States with higher revenue, lower fines, and fewer deficiencies are scored more favorably.

Staffing: Staffing-related metrics, including employee average and staff increase/decrease, are factored in. States with stable or increasing staffing levels may be considered more reliable for investment.

Certified Beds: The number of certified beds by state reflects the capacity of the facilities. A higher number of certified beds can be indicative of a stronger and more scalable operation.

Gross Patient Revenue: Gross patient revenue is considered as a financial performance indicator. Higher revenue may signify a thriving and in-demand healthcare market.

Scoring Process:

Each criterion is assigned a weight based on its perceived importance in investment decision-making.

States are scored on each criterion, and the scores are normalized to ensure a consistent scale across all metrics.

The normalized scores for each state are then multiplied by their respective weights.

The final investment score is the sum of these weighted scores for each state.

Interpretation:

Higher Investment Score: Indicates a more favorable environment for investing in skilled nursing facilities.

Lower Investment Score: Suggests potential challenges or less attractive investment opportunities.

Investors can utilize this comprehensive score to prioritize states for investment, offering a data-driven and comparative analysis to guide decision-making in the complex landscape of skilled nursing facility investments.

The Python code for the Investment Score is below.

Investment_Score.py

```

1 import pandas as pd
2
3 # Load the combined data from the output file
4 output_file_path = r'C:\Users\keros\.venv\Capstone-Project\
   output_data.xlsx'
5 combined_data = pd.read_excel(output_file_path)
6
7 # Your scoring calculations here
8 combined_data['Investment_Score'] = (
9     combined_data['Gross Patient Revenue'] / 1000 +
10    combined_data['Total Deficiencies'] * 0.5 +
11    combined_data['Fines'] * 0.8
12 )
13
14 # Sort by the Investment_Score in descending order
15 sorted_data = combined_data.sort_values(by='Investment_Score',
   ascending=False)
16
17 # Select the top and bottom 10 rows
18 top_10 = sorted_data.head(10)
19 bottom_10 = sorted_data.tail(10)
20
21 # Print the top and bottom 10 states
22 print("Top 10 States:")
23 print(top_10)
24
25 print("\nBottom 10 States:")
26 print(bottom_10)
27
28 # Concatenate the top and bottom 10 subsets
29 result_data = pd.concat([top_10, bottom_10])
30
31 # Set the path for the output Excel file
32 output_result_path = r'C:\Users\keros\.venv\Capstone-Project\
   investment_results.xlsx'
33
34 # Write the results to the output Excel file without including
   the index
35 result_data.to_excel(output_result_path, index=False)

```

```

36
37 print(f"\nTop and bottom 10 investment results saved to {
    output_result_path}")

```

Top 10 States:

	State	Facility Total	...	Gross Patient Revenue	Investment_Score
4	California	1170.0	...	15833208.0	44385.208
32	New York	606.0	...	16677396.0	19910.796
9	Florida	697.0	...	10721617.0	18212.717
36	Oklahoma	292.0	...	1726879.0	17519.379
15	Iowa	412.0	...	2507578.0	16589.278
45	Vermont	35.0	...	396837.0	15299.437
35	Ohio	946.0	...	9053725.0	14582.025
38	Pennsylvania	672.0	...	10964321.0	13493.221
25	Missouri	510.0	...	3829977.0	13168.877
13	Illinois	694.0	...	7559970.0	12576.270

[10 rows x 13 columns]

Bottom 10 States:

	State	Facility Total	...	Gross Patient Revenue	Investment_Score
41	South Dakota	98.0	...	519646.0	1636.646
29	New Hampshire	73.0	...	904564.0	1615.764
1	Alaska	20.0	...	165283.0	1489.483
12	Idaho	80.0	...	632175.0	1405.075
8	District of Columbia	17.0	...	424978.0	1295.778
31	New Mexico	68.0	...	705338.0	1282.338
7	Delaware	44.0	...	699492.0	1191.492
11	Hawaii	43.0	...	604409.0	629.809
51	NaN	NaN	...	NaN	NaN
52	NaN	NaN	...	NaN	NaN

Fig. 3. Top and Bottom 10 States for Investment.

To further provide clarity on the results, Tableau was used to provide visuals. Below is a Tableau dashboard. It shows the monthly and annual nursing home cost by state as well as the Relationship between Facility Total, Certified Bed by State and Gross Patient Revenue(GPR). The Monthly and Annual stat Cost can be used to see which state charges the most for patient care. When deciding which state would be the best to invest in, high annual cost can play a major factor. In this visual, all 50 states including Washington D.C. are shown with their state cost. Based on the data, Alaska is far and above the most expensive State while Texas has the cheapest. The bottom graph shows the margin between Facility Total and Certified Beds and GPR. The number of facilities and the beds can be compared with GPR to show the exuberant amount of revenue the industry generates as well as knowledge that facility /beds available are really low when you add in the growing baby boomer population What is Gross Patient Revenue and what factors influence it? The factors that determine GPR is not static. Several factors play a crucial role in influencing the gross patient revenue of nursing home facilities. Occupancy rates, representing the number of occupied beds, have a direct impact on revenue as higher occupancy rates lead to more billable services and increased revenue. The complexity of residents' medical conditions and care needs, known as case mix, can also affect revenue, with facilities receiving different reimbursement rates based on the level of care required for each resident. The payor mix, which includes private pay, Medicaid, and Medicare patients, influences revenue, with private pay patients contributing more revenue per day compared to Medicaid patients. The reimbursement system, especially through government programs like Medicare and Medicaid, is a critical factor, and changes in reimbursement rates or methodologies can directly impact revenue. Adequate staffing levels are essential for quality care, but labor costs can also impact a facility's financial health, emphasizing the importance of striking the right balance. Additional services such as rehabilitation and specialized therapies contribute to increased revenue, and the geographic location of a nursing home can influence local market dynamics and the ability to attract residents who can pay higher fees. A facility's reputation for providing high-quality care can attract residents willing to pay more for superior services, and regulatory compliance is essential to avoid penalties or loss of accreditation that can affect financial stability. Economic factors, such as general economic conditions, can impact the affordability of nursing home care for residents and their families, potentially leading to decreased demand during economic downturns. Effective marketing strategies and community outreach efforts also play a role in influencing the choice of a particular facility by residents and families, thereby affecting revenue. It's important to recognize the interconnected nature of these factors, as changes in one area can have ripple effects throughout the financial health of a nursing home facility. Additionally, regulatory and policy changes at the state or federal level can significantly impact reimbursement rates and overall revenue.

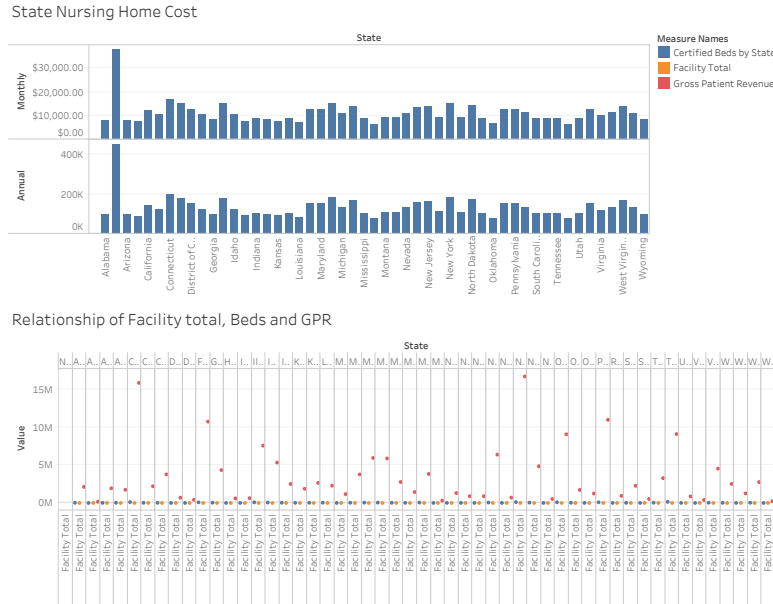


Fig. 4. Project Dashboard

7 Conclusion

Drawing insights from the presented evidence, three key observations emerge. Firstly, following the established criteria, ten states stand out as prime investment opportunities within the nursing home industry, while an additional ten states are identified as less favorable for investment. Secondly, the evidence underscores the robustness of the nursing home industry in the United States, indicating a sustained growth trajectory for the foreseeable future. This growth is substantiated by positive trends in crucial factors such as the general population and facility staffing. However, it is imperative to acknowledge that violations pose a notable concern within the industry. Consequently, for investment in these identified states to be successful, a paramount focus on facility oversight becomes essential.

8 Github Link

<https://github.com/Kells2234/Capstone-Project/blob/main/README.md>



References

1. <https://www.retirefearless.com/post/average-life-expectancy-of-a-nursing-home-resident>
2. U.S. skilled nursing facility market size, share report, 2030, <https://www.grandviewresearch.com/industry-analysis/us-skilled-nursing-facility-market>
3. (Oct 2023), https://en.wikipedia.org/wiki/Nursing_home
4. Braun, R.T., Jung, H.Y., Casalino, L.P., Myslinski, Z., Unruh, M.A.: Association of private equity investment in us nursing homes with the quality and cost of care for long-stay residents. In: JAMA Health Forum. vol. 2, pp. e213817–e213817. American Medical Association (2021)
5. Carlton, G.: Addressing the nursing home crisis: A new federal proposal makes nurses part of the solution (Oct 2023), <https://nursejournal.org/articles/addressing-nursing-home-crisis/>
6. Obando, S.: Investors are showing greater interest in nursing home acquisitions (Mar 2021), <https://www.wealthmanagement.com/seniors-housing/investors-are-showing-greater-interest-nursing-home-acquisitions>
7. Reiland, J.: Nursing home staffing shortage is a “crisis on top of a crisis” (Mar 2022), <https://skillednursingnews.com/2022/03/nursing-home-staffing-shortage-is-a-crisis-on-top-of-a-crisis/>
8. Starkey, K.B., Weech-Maldonado, R., Mor, V.: Market competition and quality of care in the nursing home industry. *Journal of health care finance* **32**(1), 67–81 (2005)
9. Wong, C.: How lucrative is the nursing home business? (Nov 2023), <https://experience.care/blog/how-lucrative-is-the-nursing-home-business/>